Exercice 1 : Zoo

Question 1

Entitée faible

Zoo(idZoo, nom)
Enclos(idEnclos, idZoo, nom, surface)

$$\text{Enclos}[\text{idZoo}] \subseteq \text{Zoo}[\text{idZoo}]$$

Cardinalité 1..1

Menu(idMenu, qteViande, qteLegume)
Famille(race, origine, ami)
Animal(idAnimal, nom, dateNaissance, race, origin, idMenu)

$$\begin{cases} \text{Animal}[\text{idMenu}] \subseteq \text{Menu}[\text{idMenu}] \\ \text{Animal}[(\text{race, origine})] \subseteq \text{Famille}[(\text{race, origine})] \end{cases}$$

Cardinalité 0..1

Animal(idAnimal, nom, dateNaissance, race, origin, idMenu)
Enclos(idEnclos, idZoo, nom, surface)

Occupe(idAnimal, idEnclos, idZoo)

$$\begin{cases} \text{Occupe}[\text{idAnimal}] \subseteq \text{Animal}[\text{idAnimal}] \\ \text{Occupe}[\text{idEnclos}] \subseteq \text{Enclos}[\text{idEnclos}] \end{cases}$$

Tables ajoutés

Ces tables ont étés ajoutés pour faire la relation parent enfant de l'Animal :
AnimalParent(idAnimalP, idAnimalE)
AnimalEnfant(idAnimalE, idAnimalP)

Question 2

```
CREATE TABLE Zoo (
    idZoo CHAR(5),
    nom VARCHAR(20),
    CONSTRAINT pk_Zoo PRIMARY KEY(idZoo));


CREATE TABLE Enclos (
    idEnclos CHAR(5), idZoo CHAR(5),
    nom VARCHAR(20), surface INTEGER,
    CONSTRAINT pk_Enclos PRIMARY KEY(idEnclos, idZoo),
    CONSTRAINT fk_Zoo FOREIGN KEY (idZoo) REFERENCES Zoo(idZoo));



CREATE TABLE Menu (
    idMenu CHAR(5),
    qteViande REAL,
    qteLegume REAL,
    CONSTRAINT pk_Menu PRIMARY KEY(idMenu));

CREATE TABLE Famille (
    race VARCHAR(20),
    origine VARCHAR(20),
    ami BOOLEAN,
    CONSTRAINT pk_Famille PRIMARY KEY(race, origine));


CREATE TABLE Animal (
    idAnimal CHAR(5),
    nom VARCHAR(20),
    dateNaissance DATE,
    race VARCHAR(20),
```

```
    origine VARCHAR(20),
    idMenu CHAR(5),
    CONSTRAINT pk_Animal PRIMARY KEY(idAnimal),
    CONSTRAINT fk_Animal_Famille FOREIGN KEY(race, origine) REFERENCES Famille(race, origine),
    CONSTRAINT fk_Animal_Menu FOREIGN KEY(idMenu) REFERENCES Menu(idMenu));

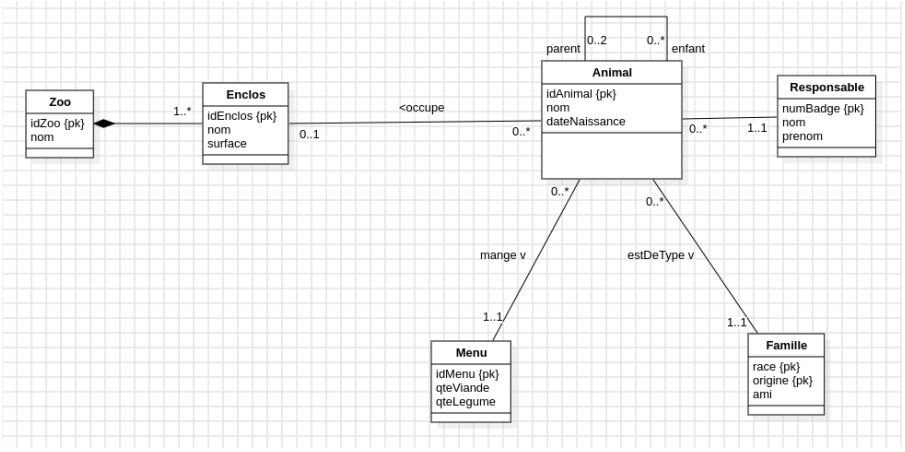

CREATE TABLE AnimalParent (
    idAnimalP CHAR(5),
    idAnimalE CHAR(5),
    CONSTRAINT pk_AnimalParent PRIMARY KEY(idAnimalP),
    CONSTRAINT fk_AnimalParent FOREIGN KEY(idAnimalP) REFERENCES Animal(idAnimal),
    CONSTRAINT fk_AnimalEnfant FOREIGN KEY(idAnimalE) REFERENCES Animal(idAnimal));


CREATE TABLE AnimalEnfant (
    idAnimalP CHAR(5),
    idAnimalE CHAR(5),
    CONSTRAINT pk_AnimalEnfant PRIMARY KEY(idAnimalE),
    CONSTRAINT fk_AnimalEnfant FOREIGN KEY(idAnimalE) REFERENCES Animal(idAnimal),
    CONSTRAINT fk_AnimalParent FOREIGN KEY(idAnimalP) REFERENCES Animal(idAnimal));




CREATE TABLE occupe (
    idAnimal CHAR(5),
    idEnclos CHAR(5),
    idZoo CHAR(5),
    CONSTRAINT pk_occupe PRIMARY KEY(idAnimal),
    CONSTRAINT fk_occupe_Animal FOREIGN KEY(idAnimal) REFERENCES animal(idAnimal),
    CONSTRAINT fk_occupe_Enclos FOREIGN KEY(idEnclos, idZoo) REFERENCES enclos(idEnclos, idZoo));
```

Question 3



Question 4

Responsable(<u>numBadge</u>, nom, prenom)
Animal(<u>idAnimal</u>, nom, dateNaissance, <u>race</u>, <u>origin</u>, <u>idMenu</u>, <u>numBadge</u>)

$$\text{Animal}[\text{numBadge}] \subseteq \text{Responsable}[\text{numBadge}]$$

Queston 5

```
CREATE TABLE Responsable (
    numBadge CHAR(5),
    nom VARCHAR(20),
    prenom VARCHAR(20),
    CONSTRAINT pk_responsable PRIMARY KEY(numBadge));

ALTER TABLE Animal ADD numBadge CHAR(5);

ALTER TABLE Animal ADD CONSTRAINT fk_Animal_Responsable FOREIGN KEY(numBadge) REFERENCES Responsable(numBadge);
```

Question 6

```sql
INSERT INTO Zoo VALUES ('Z1', 'Zoo de Valence');

INSERT INTO Enclos VALUES ('E1','Z1', 'Royaume des Lions', 2300);
INSERT INTO Enclos VALUES ('E2','Z1', 'Savane', 1200);

INSERT INTO Menu VALUES ('M1', 7.5, 0);
INSERT INTO Menu VALUES ('M2', 0, 40 );

INSERT INTO  Responsable VALUES('12345','DURAND', 'Olivier');
INSERT INTO  Responsable VALUES('56789','DUPOND', 'Nathalie');
```